Logarithmic Scale Weighted Cross-Entropy

for Class-Imbalanced Learning

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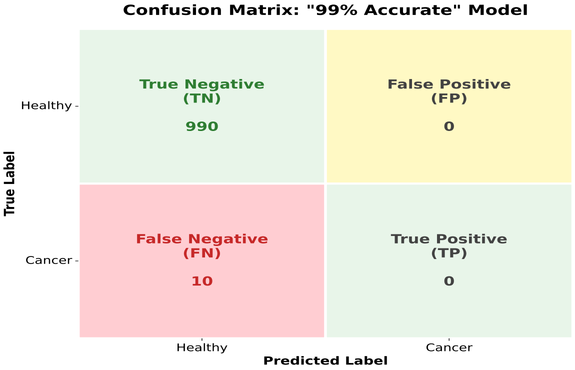
Latent Majority Bias in Imbalanced Data

- Class imbalance: Common in healthcare, finance, manufacturing.
- Models on imbalanced data: High accuracy by ignoring minorities $\rightarrow$ deceptive.
- Fatal risks: Missed rare diseases, undetected fraud, overlooked defects.



Common Approaches to Class Imbalance

1. Data Resampling

- **What it is:** Modifies the training set distribution to achieve a better balance.
- **Examples:**
  - Oversampling (SMOTE, ADASYN)
  - Undersampling (Tomek Links, ENN)

2. Cost-Sensitive Learning

- **What it is:** Adjusts the algorithm to assign a higher penalty for misclassifying minority samples.
- **Examples:**
  - Weighted Cross-Entropy (WCE)
  - Focal Loss, Class-Balanced Loss

Standard Cross-Entropy:

$$\mathcal{L}_{CE} = -\sum_{i=1}^{N} y_i \log(p_i)$$

Problem: The total loss is dominated by the majority class, and the gradient signal from the minority class is overwhelmed.

$\rightarrow$ *Focus on* **Cost-Sensitive Learning**.

Existing Methods (I): Inverse Weighting Schemes

The simplest cost-sensitive method is to assign a weight w_i to the standard CE loss:

$$\mathcal{L}_{WCE} = -\sum_{i=1}^{N} w_i \cdot y_i \log(p_i)$$

Weighted Cross-Entropy (WCE) & Inverse Sqrt WCE

- **Core Idea:** Weights are inversely proportional to class frequency, sometimes softened.
- **Weight Formula** (n_i: samples in class i)**:**

$$w_i \propto \frac{1}{n_i} \quad \text{or} \quad w_i \propto \frac{1}{\sqrt{n_i}}$$

Existing Methods (II): Advanced Losses

1. Focal Loss

- **Core Idea:** Focuses on sample difficulty instead of class frequency.
- Down-weights the loss of "easy" (often majority) samples.
- **Formula** (p_t: prob. of true class):

$$\mathcal{L}_{FL} = -\alpha_t (1 - p_t)^{\gamma} \log(p_t)$$

- **Key Hyperparameter:**
  - γ controls the focusing rate.

2. Class-Balanced (CB) Loss

- **Core Idea:** A principled re-weighting based on the "effective number of samples".
- **Weight Formula ($N_i(n_i, \beta)$):**

$$w_i = \frac{1 - \beta}{1 - \beta^{n_i}}$$

$$\mathcal{L}_{CB} = -\sum w_i \cdot \mathcal{L}_{CE}(y_i, p_i)$$

- **Key Hyperparameter:**
  - β controls the curve.

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Proposed Methodology (I): The Core Idea

Motivation: Instability of WCE

Standard WCE weight:

$$w_i = \frac{N_{total}}{n_i} \propto \frac{1}{n_i}$$

Critical Issue

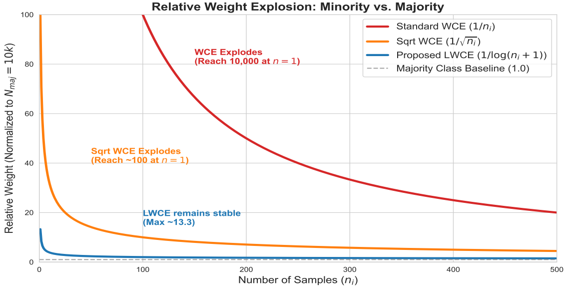
Severe imbalance ($n_i \to$ small) causes w_i to explode.

- Numerical instability.
- Severe overfitting to minority

Solution: Log-Weighted CE (LWCE)

We use log-scaling to tame the weights:

$$w_i \propto \frac{1}{\log(n_i + 1)}$$



Proposed Methodology (II): Power-Scaled LWCE (PLWCE)

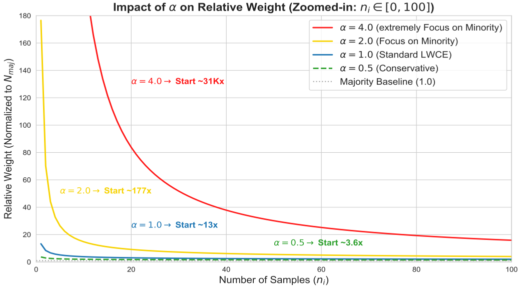
Generalization: Power-Scaled Adaptive CE (PLWCE)

We introduce α to flexibly control the focusing strength:

$$w_i \propto \frac{1}{(\log(n_i + 1))^{\alpha}}$$

Alpha (α)	**Role & Effect**
$\alpha = 1.0$	Balanced LWCE
$\alpha > 1.0$	Aggressive focus on minority
$\alpha < 1.0$	Conservative weighting

Table: Role of the α parameter



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Experimental Setup (I): Datasets Model

Table: Benchmark Datasets (Sorted by IR)

Dataset	# Classes	Imbalance Ratio (IR)
Credit Card Fraud	2	587:1
APS Failure	2	59:1
Page Blocks	5	175:1
Yeast	10	93:1
... (Total 11)	...	...
German Credit	2	2.3:1

Base Model: XGBoost

XGBoost with linear booster (‘gblinear‘) as a robust baseline.

- Handles both binary and multi-class tasks.
- Efficient for large-scale tabular data.

Experimental Setup (II): Metrics Optimization

Evaluation Metrics

Standard accuracy is misleading for imbalanced data. We focus on:

- **F1-Score:** Harmonic mean of precision and recall, averaged across classes.
- **PR-AUC:** Area under the Precision-Recall curve, highly sensitive to minority class performance.

Hyperparameter Optimization

We used **Optuna** for automated hyperparameter tuning.

- **Search Space:**
  - `learning_rate`, `n_estimators`
  - Regularization: `lambda` (L2), `alpha` (L1)
  - Loss-specific: γ (Focal), β (CB), α (PS-WCE, PLWCE)
- **Protocol:** 3-fold CV, 100 trials per dataset.

Full Experimental Results (I)

Credit Card Fraud

Loss	F1	PR
CE	0.722	**0.711**
WCE	0.796	0.698
WCE(sq)	0.797	0.700
PS-WCE	**0.799**	0.701
Focal	0.757	**0.715**
CB	**0.801**	0.704
LWCE	0.747	**0.716**
PLWCE	**0.806**	0.701

Aps Failure

Loss	F1	PR
CE	0.759	0.812
WCE	0.648	0.813
WCE(sq)	0.761	0.825
PS-WCE	**0.776**	0.825
Focal	**0.774**	**0.826**
CB	0.756	**0.827**
LWCE	0.765	0.824
PLWCE	**0.773**	**0.828**

Secom

Loss	F1	PR
CE	0.167	0.141
WCE	**0.276**	**0.167**
WCE(sq)	0.172	0.137
PS-WCE	0.189	0.136
Focal	0.145	0.139
CB	0.128	0.142
LWCE	**0.222**	**0.177**
PLWCE	**0.213**	**0.171**

Bank Marketing

Loss	F1	PR
CE	0.456	**0.548**
WCE	0.562	0.539
WCE(sq)	**0.574**	0.545
PS-WCE	**0.573**	0.545
Focal	**0.567**	**0.546**
CB	0.474	0.503
LWCE	0.489	**0.548**
PLWCE	0.558	**0.546**

Credit Card Default

Loss	F1	PR
CE	0.357	**0.498**
WCE	**0.501**	0.449
WCE(sq)	**0.496**	**0.483**
PS-WCE	0.483	0.479
Focal	**0.510**	**0.48**
CB	0.487	0.472
LWCE	0.488	0.473
PLWCE	0.484	0.479

Telco Churn

Loss	F1	PR
CE	0.593	0.628
WCE	0.623	**0.643**
WCE(sq)	**0.629**	0.629
PS-WCE	0.624	**0.641**
Focal	0.623	**0.635**
CB	**0.630**	0.626
LWCE	**0.629**	0.625
PLWCE	**0.629**	0.624

Full Experimental Results (II)

German Credit

Loss	F1	PR
CE	0.585	0.640
WCE	**0.630**	0.631
WCE(sq)	0.615	**0.644**
PS-WCE	0.602	0.639
Focal	0.601	**0.641**
CB	**0.626**	**0.641**
LWCE	0.591	**0.641**
PLWCE	**0.628**	**0.642**

Page Blocks (Multi)

Loss	F1	PR
CE	0.829	**0.885**
WCE	0.006	0.200
WCE(sq)	**0.835**	0.883
PS-WCE	0.813	0.882
Focal	0.792	0.843
CB	**0.838**	0.883
LWCE	0.815	**0.892**
PLWCE	**0.834**	**0.893**

Yeast (Multi)

Loss	F1	PR
CE	**0.577**	**0.597**
WCE	0.447	0.570
WCE(sq)	0.521	0.593
PS-WCE	**0.533**	0.593
Focal	0.462	0.514
CB	0.518	0.592
LWCE	**0.541**	**0.598**
PLWCE	0.528	**0.594**

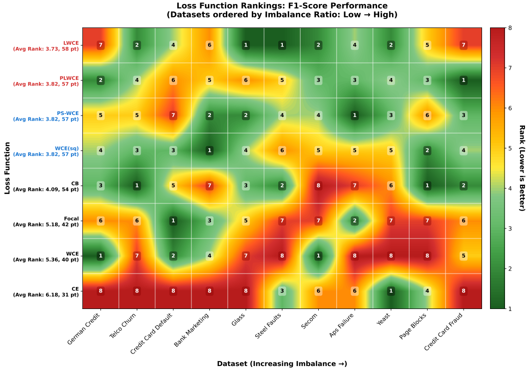
Steel Faults (Multi)

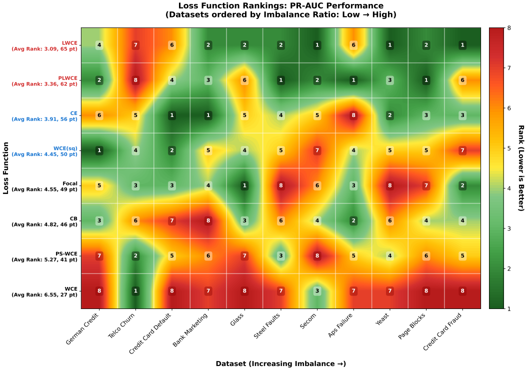
Loss	F1	PR
CE	**0.706**	**0.726**
WCE	0.628	0.721
WCE(sq)	0.691	**0.725**
PS-WCE	0.698	**0.726**
Focal	0.667	0.707
CB	**0.706**	**0.725**
LWCE	**0.710**	**0.726**
PLWCE	0.696	**0.726**

Glass (Multi)

Loss	F1	PR
CE	0.661	0.731
WCE	0.668	0.695
WCE(sq)	0.707	0.735
PS-WCE	**0.719**	0.705
Focal	0.698	**0.744**
CB	**0.710**	**0.736**
LWCE	**0.736**	**0.740**
PLWCE	0.678	0.719

Overall Performance Summary: Heatmap





Key Finding: Robustness

LWCE & **PLWCE** are consistently top-ranked (Y-axis). The green colors in the **top-left region** confirm their robustness on datasets with extreme imbalance.

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Discussion (I): The Benefit of Tuning α

Observation from Results

- The results show that LWCE ($\alpha = 1.0$) is already a very strong and robust baseline, often outperforming SOTA methods.
- However, in 5 out of 11 datasets, PLWCE (tuning $\alpha > 1.0$) achieved an even higher F1-Score than LWCE.

Conclusion: Alpha as a "Performance Booster"

α acts as a data-specific "booster":

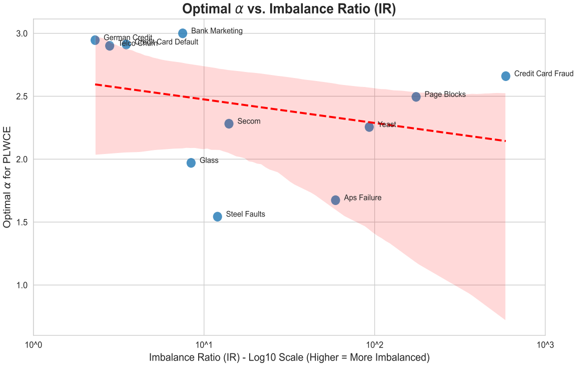
- LWCE ($\alpha = 1.0$) is sufficient and provides top-tier performance.
- For some extremely imbalanced datasets, tuning $\alpha > 1$ allows PLWCE to find a better optimum, boosting **both F1 and PR-AUC** scores.

$\rightarrow$ *PLWCE generalizes LWCE, providing an optional tuning parameter for achieving the best possible performance in critical scenarios.*

Discussion (II): Difficulty of Isolating α

Challenge: Isolating α vs. IR

- **Ideal Experiment:** Fix all model parameters (LR, L1/L2, etc.) across all 11 datasets, and tune only α.
- **Problem:** A single set of fixed parameters failed. It was too sub-optimal for diverse datasets (e.g., `credit_card_fraud`), causing poor metrics and rendering the entire imbalanced data analysis meaningless under these conditions



- Shows a weak positive trend: higher imbalance (larger IR) seems to prefer a larger α ($\alpha > 1$).

Future Work (I): Model & Weighting Expansion

1. Application to Other Models

- **Tree-based Models:** Validate LWCE/PLWCE on non-linear ‘gbtree‘ models (standard XGBoost, LightGBM).
- **Deep Learning:** Apply to CNNs/Transformers for highly imbalanced tasks like object detection or medical segmentation.

2. Combining with Dynamic Sample Weighting

- Our method (PLWCE) uses static, class-level weights.
- **Hypothesis:** We can combine this with dynamic, sample-level methods (like Focal Loss or GHM) that focus on "easy" vs. "hard" samples.
- **Goal:** Investigate a hybrid loss that benefits from both stable class-based re-weighting (our α) and dynamic sample-level focus (like Focal's γ).

Future Work (II): Automated Alpha (α) Tuning

3. Automated Alpha (α) Tuning

- As discussed (P18), isolating α's effect is challenging due to other interacting hyperparameters.
- **Controlled Experiment:**
  - Use a large dataset (e.g., `credit_card_fraud`).
  - Fix minority samples and sub-sample the majority class to create 5+ datasets with controlled IRs (e.g., 5:1, 10:1, 100:1).
- **Goal:** Find a clear correlation between IR and the optimal α by running alpha-only tuning on these controlled datasets.
- **Ultimate Aim:** Develop a heuristic or formula to set α automatically, eliminating the need for expensive trial-and-error tuning.

Conclusion

Key Contributions

1. **Identified Instability:** We highlighted the numerical instability of standard WCE ($1/n_i$) in extreme imbalance scenarios.
2. **Proposed LWCE:** We proposed **LWCE** ($1/\log(n_i + 1)$) as a simple, stable, and parameter-free solution that fundamentally tames weight explosion.
3. **Proposed PLWCE:** We generalized this into **PLWCE** ($1/(\log n_i)^\alpha$), demonstrating how α acts as a performance booster for extremely imbalanced datasets.
4. **Validated Performance:** Our methods proved highly robust and competitive, consistently ranking in the Top 2 across 11 diverse datasets against strong baselines.

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Thank You for Your Attention!

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